

Rechnernetze - Challenge Lab Preparation

Answer all provided questions within a separate document.

1. Create the basic configuration for all 3 routers:

Questions: Please examine the running configuration of R1. Explain which of the three types of passwords is responsible for which method of access.

Why is the enable secret password not visible in clear text? What method was used to secure the password (pay attention to the value of "5" after enable secret).

Examine the routing tables on each router. **Why can R1 not ping R3?**

2. Configure the S1 switch:

Questions: What are the advantages of using VLANs? What are the differences between a collision and a broadcast domain? What is the difference between an access port and a trunk port?

3. Configure the Inter-VLAN routing on R1:

Questions: What is the purpose of the gateway information, and which networking device normally acts as the gateway? Why is inter-VLAN routing necessary for hosts on different VLANs to communicate with each other?

4. Configure DHCP on R1:

Questions: Why are several addresses excluded from being assigned by DHCP? What are typical devices that would not be configured by DHCP? What happens with new devices when all IP address have been used once? What are lease times?

5. Routing configuration:

Questions: Examining the routing tables of all routers, which networks are present in the routing table and how are they connected?

What is the difference between static and dynamic routing? What is the purpose of a routing protocol?

6. Analyze packets in simulation mode:

Questions: What is the difference between TCP and UDP and which is used by RIP? Which port does RIP use? What is the difference between RIP version 1 and version 2 (check the destination addresses)? Display the outbound PDU details and take a screenshot of the RIP information. **Explain how routers exchange information and how they use the information within the RIP packets to build the routing table.**

Reset the simulation and use the command prompt on a PC and delete the arp cache with "arp -d". Then ping the FA0/0 interface of R3. **What is the purpose of ARP and why was an ARP request sent out by the PC? Who responds to this request?**

Reset the simulation again and use to the command prompt of the PC again to traceroute (tracert) to the FA 0/0 interface of R3. Display the outbound PDU details of each ICMP packet the PC is sending. **Inspect the "time to live" value (TTL) within the IP header and explain how traceroute is able to display all routing hops on its way to the target. What happens with the TTL value when passing each router?**